

The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach — Milestone 3

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Where we left off

Major feedbacks:

- 1 **Stronger justification that the selected donors satisfy SUTVA** — i.e. that no donor is itself contaminated by the treatment.
- 2 **Economic interpretation of why the models behaved as they did** during the treatment period
- 3 **Explore a naive counterfactual** — show what the SCM machinery buys us over simpler baselines.

Re suggested reading **Ham & Miratrix**: it concerns pre-trend identification for *matching / difference-in-differences* designs, not synthetic control on a price series, so its specific machinery does not transfer to our setting

Comment 1 — donor selection and SUTVA

The selection question, asked of every candidate:

Does the asset co-move with Brent through common global factors, yet stay off the oil-supply / disruption path?

Yes to both → include. On the oil-supply / disruption path (chokepoint or Russia/Ukraine trade) → exclude (SUTVA).

Two contamination channels to screen out:

- **The oil-supply path** — energy complex (WTI, gas, refined products) and petrocurrencies (CAD, NOK, BRL, ...). Categorically on the oil-supply path → never candidates.
- **The disruption itself** — Persian-Gulf routing (Hormuz) or Russia/Ukraine trade (Russia 2022). Screened donor-by-donor, category by category →

Donor selection — Metals & Agriculturals

Metals — Silver, Platinum, Gold

- **Include:** real-rate / safe-haven / dollar channel — a monetary co-movement with Brent, off the oil-supply path.
- **Exclude:** Copper, IronOre (Russia metal supply $\sim 4\text{--}5\%$ world), Palladium (Russia $\approx 40\%$ world supply, spiked Mar-2022).

Agriculturals — Coffee, Sugar, LiveCattle

- **Include:** global trade / demand and the dollar; producers outside the conflict zones (Brazil, India, US).
- **Exclude:** Wheat, Corn (Russia+Ukraine $\approx 30\%$ / 13% world exports), Soybeans (grain cross-elasticity), Cotton (oil $\rightarrow$ polyester substitution).

Donor selection — Equities & FX

Equities — SP500, Nikkei

- **Include:** global growth and the equity risk premium — the demand channel shared with oil; no Russia or Gulf-energy weight.
- **Exclude:** EM_Eq (Russia index weight); energy-heavy DAX / FTSE / TSX-60 never enter.

FX — AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN

- **Include:** safe-haven (JPY, CHF), EM-growth (ZAR, MXN, INR, KRW), China-demand (CNY, AUD) — dollar and growth factors.
- **Exclude:** EUR, DXY (EU energy-crisis / dollar on the invasion), GBP (UK gas); petrocurrencies (CAD, NOK, BRL, ...) never enter.

Donor selection — Rates / credit & Volatility

Rates / credit — TLT, HYG

- **Include:** TLT (US duration) and HYG (credit-risk premium) — the discount-rate / risk side of the global factor, not oil-linked.
- **Exclude:** US10Y (Russia: inflation-expectations / flight-to-quality loads it onto the treatment).

Volatility — VIX

- **Include:** the single risk-on / risk-off switch — the fear premium that also lifts oil in stress.
- **Exclude:** none — one volatility proxy, kept deliberately to avoid over-weighting the risk channel.

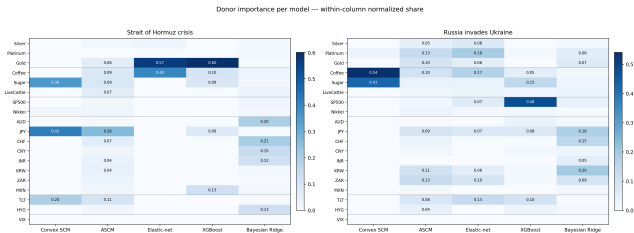
Comment 1 — SUTVA test: structural breakpoints at T_0

A clean donor should show *no* structural break at the event. Two tests: **Bai–Perron** (level breaks in the donor–Brent relationship) (Bai & Perron 2003); **HLT** (trend break, $I(0)/I(1)$ -robust) (Harvey, Leybourne & Taylor 2009).

Break near T_0 (of 19)	Hormuz 2026	Russia 2022
Bai–Perron	0 / 19	1 / 19 (VIX)
HLT trend-break	0 / 19	0 / 19

- **No excluded donor breaks** at either T_0 .
- Lone flag: **VIX** (included) — the generic risk-off spike at the invasion, i.e. intended common-factor co-movement, not Russia-specific contamination.

Comment 2 — how the donor weights drive model behaviour



Reading the heatmap.

Column-normalised donor importance per model (convex weight / $|\hat{\beta}_j|$ / feature importance).

Hormuz. Weight concentrates on *safe-haven metals* (Gold, Platinum) and *JPY / VIX*. These rose on generic risk but *not* on the oil-specific shock → synthetic undershoots actual Brent → large *positive* gap.

Russia — divergence is donor-driven.

The models split because they lean on *different* donors: Convex/ASCM concentrate on Coffee & Platinum ($w \approx 0.5-0.65$), XGBoost leans SP500 (~ 0.48), while Bayesian Ridge blows up on JPY & KRW ($w \approx 1.8-2.4$). far

Comment 2 — the contamination bias is signed, and it makes us conservative

The critique. If high oil prices *crowd out demand*, the donors that proxy global activity are themselves pushed *down* by the treatment → the synthetic is dragged down with them → residual contamination in the gap. **So: which way does the bias go, and how big?**

We sign it. For each linear model we decompose the gap into a clean component and a donor-contamination component (net bias) over the full post-window:

Model	Hormuz net bias	Russia net bias
Convex SCM	−1.3% (conservative)	+21.3%
ASCM	−2.3% (conservative)	+11.8%
Elastic-net	−2.6% (conservative)	+19.5%
Bayesian Ridge	−7.3% (conservative)	−7.5%
XGBoost	n/a (nonlinear)	n/a (nonlinear)

Source: `data/validation/contamination_bias_decomposition.csv`.

- **Hormuz: bias is negative across all linear models** → the demand-destruction channel makes the SCM *under-state* the premium. The headline **~59%** is therefore a **lower bound**, not an inflated one.

Statistical test 1 — in-space placebo

Procedure (Abadie 2010): refit the SCM treating each donor as treated; rank Brent's post/pre RMSPE ratio against the $N=20$ units (Brent + 19 donors). **Permutation** $p = \text{rank} / N$; floor $p = 1/20 = 0.05 \rightarrow$ rank 1 is the strongest signal the 19-donor pool can produce.

Model	Hormuz 2026	Russia 2022
	rank $\rightarrow p$	rank $\rightarrow p$
Convex SCM	1 $\rightarrow$ 0.05	10 $\rightarrow$ 0.50
ASCM	1 $\rightarrow$ 0.05	12 $\rightarrow$ 0.60
Elastic-net	1 $\rightarrow$ 0.05	8 $\rightarrow$ 0.40
XGBoost	1 $\rightarrow$ 0.05	4 $\rightarrow$ 0.20
Bayesian Ridge	1 $\rightarrow$ 0.05	12 $\rightarrow$ 0.60

Source: data/validation/inference_inspace_brent_summary.csv (19-donor shared pool).

- **Hormuz:** Brent ranks 1 for **all five models** $\rightarrow$ its break beats every placebo; strongest signal the pool allows.
- **Russia:** mid-pack (rank 8–12) for the linear models; XGBoost is best at rank 4 ($p=0.20$) but still not significant — consistent with Russia as a magnitude check, not a clean inferential case.

Statistical test 2 — leave-one-out (LOO)

Procedure. Drop one influential donor ($w > 5\%$) at a time, refit, recompute the ATT. Report the **range** (max — min, pp) across the leave-one-donor-out refits — a small range means no single donor drives the result.

Model	Hormuz LOO range (pp)	Russia LOO range (pp)
Convex SCM	8.2	26.8
ASCM	5.7	19.1
Elastic-net	6.7	18.8
XGBoost	6.1	13.8
Bayesian Ridge	10.4	27.3 [†]

Source: `data/validation/inference_loo_{event}_{model}.csv`.

- **Hormuz: tight** (~6–10 pp on a ~59% effect) — no single donor flips the result; ASCM tightest.
- **Russia: wide** (14–27 pp). Bayesian Ridge ([†] range crosses zero, –21.9 to +5.4) is LOO-fragile, consistent with its degenerate fit; XGBoost is the most stable. Reinforces Russia as a magnitude check, not the headline.

Comment 3 — naive counterfactual baselines

Question. What does the donor-informed SCM buy us over baselines that ignore the donor pool entirely? We compare the ensemble-median SCM premium against four naive counterfactuals at the full post-window.

Counterfactual	Hormuz %	Russia %
SCM ensemble (donor-informed)	58.9	22.0
Random walk — flat	51.9	8.7
Random walk — drift	52.6	−6.9
Linear trend	73.8	−4.2
Interrupted time series (ITS)	73.8	−4.2

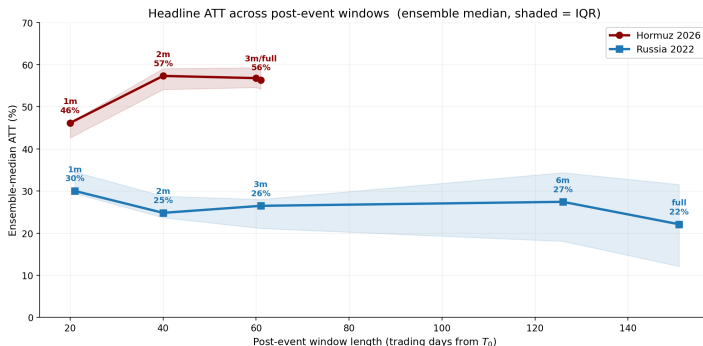
Source: `data/validation/naive_baseline_comparison.csv` (full-window horizon).

Reading.

- **Random walks understate** the premium — they freeze the counterfactual at the last pre-event level and miss the donor-implied path the rest of the market was on.
- **Linear trend / ITS overstate** Hormuz (74%) and *flip sign* on Russia (−4%) — naive extrapolation of a pre-trend is spurious when the donor factor itself turned.
- The SCM sits **between** the two failure modes: anchored to a *data-driven* counterfactual that tracks the global factor without freezing or blindly extrapolating it.

Summary — headline ATT across post-event windows

- 1 **SUTVA** — audit (oil-supply / disruption path), corroborated by the breakpoint test.
- 2 **Model behaviour** — weight on risk donors explains the gap; contamination bias signed (Hormuz conservative).
- 3 **Naive baselines** — random walks understate, trend / ITS over-extrapolate; the SCM sits between.



Hormuz builds to **~59%** by month 2–3 and holds (IQR [54, 60]); **Russia** peaks **~31%** (1m) and settles **~22%**

Thank you

Questions and discussion

github.com/ElvisCasco/Brent_Analysis